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SPPU M.Sc. Data Science — Free Study Material

Big Data

DS-560-MJ

Sample / Practice Question Paper

Semester II · Elective · Practice Paper

& This is a SAMPLE / PRACTICE question paper prepared for revision from the SPPU M.Sc. Data Science syllabus. It is NOT an official or past university examination paper. Use it only to test your preparation.

Time: 3 Hours

Maximum Marks: 35

Instructions: (1) All questions carry equal weightage where indicated. (2) Figures to the right indicate full marks. (3) Draw neat diagrams / write code wherever necessary.

Q1. Answer any FIVE of the following. [5 × 2 = 10]

- a) State the characteristics of Big Data.
- b) What is HDFS?
- c) What is MapReduce?
- d) Name two Hadoop ecosystem components.
- e) Why is Spark faster than MapReduce?
- f) What is Hive?
- g) What is a NoSQL database?

Q2. Answer the following. [10]

A) Explain the characteristics, sources and applications of Big Data.

OR

B) Explain Hadoop architecture and its components (HDFS, MapReduce).

Q3. Answer the following. [10]

A) Explain data preprocessing and real-time analytics in Big Data.

OR

B) Explain the Hadoop ecosystem: Pig, Hive, HBase, Mahout and Spark.

Q4. Answer the following. [5]

A) Explain Big Data visualization tools and best practices.

OR

B) Explain scalability and performance optimization techniques.

